

Leakage-Safe Fire & Smoke Detection with YOLOv5

A clean evaluation split exposed a failing legacy checkpoint - retraining recovered independent-test mAP@0.5 from 0.084 to 0.755.

Why this project matters

Discarded cigarettes, smouldering material and delayed smoke recognition can turn small ignition events into major incidents. AI Fireguard investigates camera-based detection that can support earlier alerts and later connect to evacuation planning.

Detection targets: CIGARETTE / FLAME / SMOKE

The audit changed the conclusion

1,511 / 188 / 188
train validation independent test

- 1,887 cleaned image-label pairs; backgrounds retained as negative samples.
- 0 source-name, exact SHA-256 or perceptual-hash overlaps across splits.
- Class 1 corrected from face-cigarette-smoking to flame.



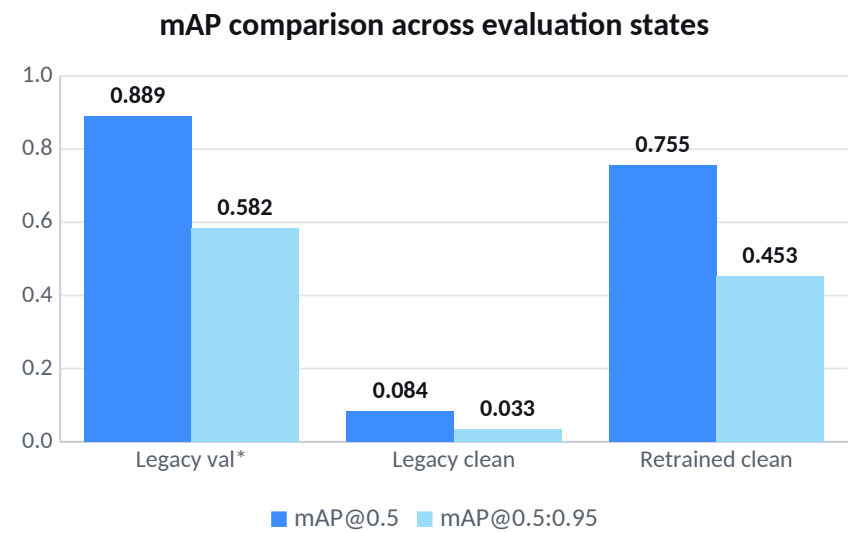
Reproducible local training

YOLOv5s | 45 epochs | 640 x 640
Batch 8 | Apple MPS | 2 workers | seed 0
Initialization: legacy best.pt; all layers unfrozen

2 h 04 min training | 0 swaps | 14 MB model

A fair test exposed the legacy gap

The legacy run looked strong on its own validation artifact, but the same checkpoint collapsed on the non-overlapping clean test. The drop is diagnostic evidence of evaluation leakage and label mismatch - not a model regression.



*Legacy validation is not directly comparable: old split and old class definition.

0.755

independent test mAP@0.5

0.453

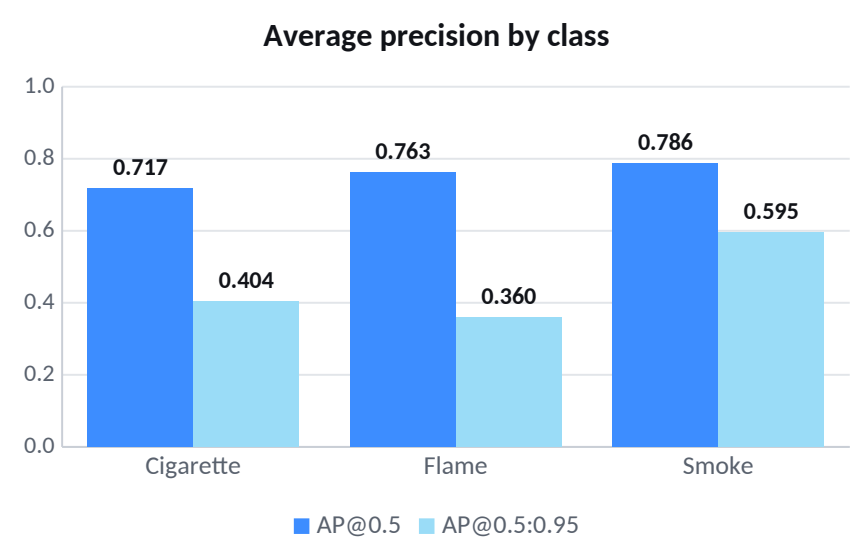
independent test mAP@0.5:0.95

Retraining recovered 85% of the legacy artifact's mAP@0.5 while using a genuinely independent test split.

Independent test summary

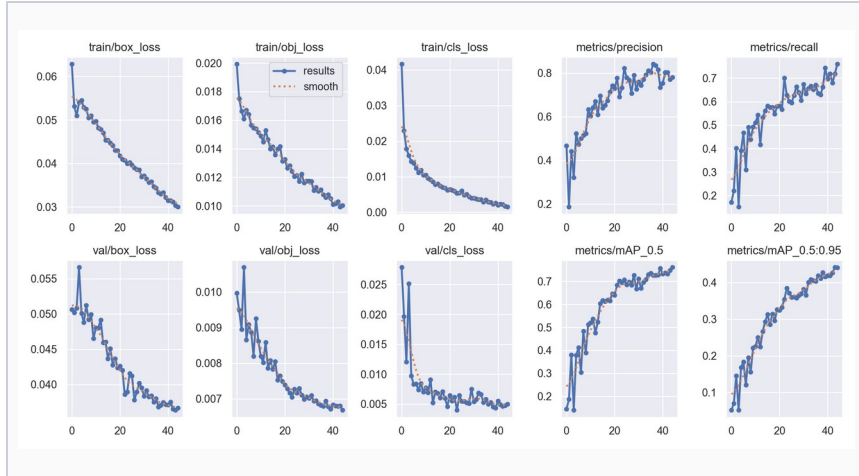
Precision 0.841 Recall 0.680
188 images | 171 objects | 11.3 ms/image at batch 8

Performance is strongest on smoke



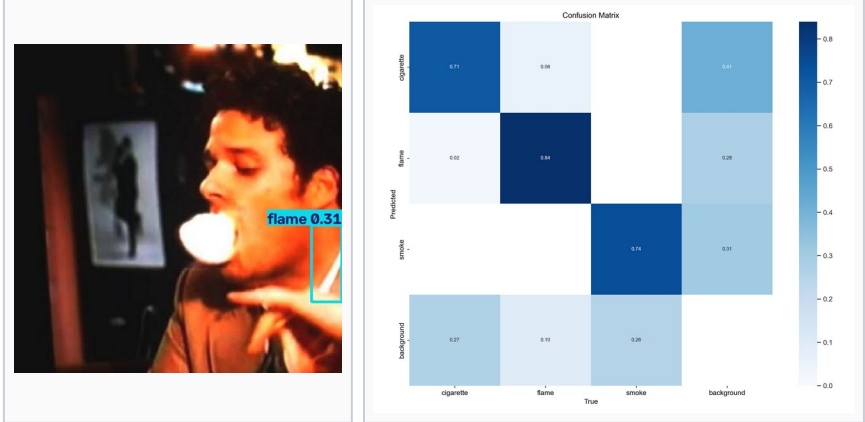
Class	Precision	Recall	Instances
Cigarette	0.794	0.634	82
Flame	0.812	0.740	50
Smoke	0.919	0.667	39

Training remained stable



Losses declined while recall and both mAP measures continued to rise through the final epochs.

Qualitative evidence



Left: single-image flame detection (0.31). Right: normalized clean-test confusion matrix.

What the errors mean

- Smoke leads: precision 0.919 and AP@0.5:0.95 0.595.
- Cigarette recall remains 0.634: small targets and hand occlusion still matter.
- Flame recall is 0.740, but AP@0.5:0.95 is 0.360: localization needs improvement.
- Missed targets dominate more than cross-class confusion on the clean test.

Deployment and next steps

- Add small-cigarette, low-light, occlusion, fog and steam hard negatives.
- Calibrate confidence per class and test temporal consistency on video.
- Validate on unseen buildings before safety-critical deployment.
- Connect verified alerts to the escape-route planning prototype.

Trustworthy splitting changed the story: the detector now generalizes on a clean test instead of only scoring well on its own validation data.